

1. Project Rationale

This proposal is for an **Integrating Research and Practice** project, *Collaborative Research: Cultivating Tomorrow's Innovators Through Exploring Planetary Images with Artificial Intelligence*

1.1 Introduction and Goals

We propose to design, implement, and research an out-of-school interdisciplinary STEM learning experience, *Knox Innovation Lab*, integrating Planetary Science, big data, creative works, and Generative Artificial Intelligence (AI). Our goal is to generate enthusiasm for STEM and inspire creativity among a diverse group of high school youths through engaging youths in creating scientifically valid and visually appealing artworks, digital stories and web applications (apps) by leveraging leading-edge AI tools, and to develop youths' science identities through a sense of belonging to a greater scientific community which will in turn, motivate them to pursue STEM interests and careers. We will employ a mix of online and in-site activities and programs through the collaborative creation and dissemination of youths' creative works in the context of East Tennessee (TN), a region in the heart of Appalachia marked by persistent challenges in providing youths with substantial learning experiences in Planetary Science, Computer Science, and emerging AI (Mehra, et al., 2010; Gibbons et al., 2020). To achieve our goal, we have brought together a diverse team of expertise, including Planetary Scientists (Moersch, Dygert and Huang), AI researchers (Hong, Maroulas and Wang), a Geospatial researcher (Wu), a learning scientist (Bass), outreach specialists (Robinson, Reeves and Al-Aseer), an informal STEM educator (Wang), and a core set of partners who have the capacity and experiences to implement and sustain the program over time including the American Museum of Science and Energy (AMSE) and the University-Assisted Community Schools (UACS). Together, we will:

1. Develop and implement *Knox Innovation Lab*, an out-of-school interdisciplinary STEM learning experience, integrating Planetary Science, big data, creative works, and AI,
2. Create the Planet+AI Art Challenge, Digital Storytelling Challenge and App Challenge which engage youths in creating scientifically valid and visually appealing artworks, digital stories and web apps by leveraging open planetary images and leading-edge AI tools,
3. Recruit ~120 youths to participate in the Knox Innovation Lab,
4. Build a Community of Practice (CoP) to facilitate youths' STEM learning and creative expression,
5. Implement a three-week Summer Institute integrating STEM into creative work productions,
6. Implement after-school webinar/workshop series in Fall and Spring semesters by invited Planetary Scientists, AI researchers, experts in creative arts, industry speakers, and project staff,
7. Showcase youths' scientifically valid artworks and technology via online art exhibits, physical art exhibits at partner museums and virtual Innovation Conference, and disseminating youths' digital stories through Medium, local NPR and East TN PBS,
8. Implement a one-week Professional Development (PD) program in the final year to support educators in either formal or informal learning settings,
9. Create a set of curriculum materials that will inform other informal STEM programs to integrate AI with other disciplines,
10. Conduct rigorous research and evaluation of Knox Innovation Lab's impact on youths' STEM learning and identity throughout participation in the project, and
11. Contribute to the informal STEM learning community and the larger science education community by better understanding how and in what ways to implement an interdisciplinary STEM program that incorporates AI with other STEM disciplines.

1.2 Building on Our Pilot Study HiRISE+AI

Our proposed project builds on PI Wang's current NASA-funded one-year [2/1/2022-1/31/2023] pilot study *HiRISE+AI* (See #1 in Reference Section) in which we develop and implement an out-of-school interdisciplinary learning experience for underrepresented high school youths in southern Nevada, integrating Planetary Science and AI. Nearly 20 underrepresented youths with no prior knowledge and skills in coding or AI participate in our project, including 2 from the Knoxville area (We have recently branched out our program to the youths in the Knoxville area). The goal of HiRISE+AI is to address youths' need to make sense of emerging technologies, particularly the "AI" appearing on TV news and in our everyday lives through engaging youths in creating meaningful aesthetic artworks with HiRISE images and leading-edge AI tools themed on planetary explorations and discoveries. In order to provide a

developmentally appropriate HiRISE+AI curriculum for high school youths, we utilize an application-focused approach. We use Arnaud Bodin's *Python in High School* and a heavily revised version of Coursera's Convolutional Neural Networks as our essential learning materials. AI-based text-to-image systems generate synthetic images by algorithmically detecting features in large volumes of data. As Yann LeCun states in 2022 that the way forward in AI is still based on convolutional neural networks-based Deep Learning; thus, we design the progression of our HiRISE+AI curriculum as the following: We firstly introduce HiRISE image data, Python basics and then how convolutional neural networks automatically learn features from input dataset through multiple layers and backpropagation. After youths develop an understanding about convolutional neural networks, we start to introduce generative adversarial networks through actually training the ProjectedGAN with custom HiRISE dataset. Visual inspection of generated synthetic images during the training ProjectedGAN gives youths a concrete experience about how an AI model generates synthetic images. Lastly, we start to show youths how to craft prompts to generate images with text-to-image systems (See Table 1 below).

Also, we introduce current planetary explorations and leading-edge AI tools by invited speakers and conduct hands-on workshops on implementation of CNN models, training ProjectedGAN, and Generative AI models on Google Colab by utilizing Participatory Live Coding (Nederbragt et al., 2020) and the "use-modify-create" framework (e.g., Lytle et al., 2019) throughout the project. Our instructors write and narrate Python code out loud on Colab and invite youths to join them by writing and executing the same code. Youths are given Colab notebooks and easy-to-follow instructions in order to reduce cognitive load for beginners. Our youths are very excited that many of the AI-generated images have distinctive qualities that almost look like real images mimicking specific styles and they offer us endless possibilities. The last activity of the pilot study is to engage youths to create artworks themed on Mars Sample Return leveraging text-to-image systems. Just like open source software all these new AI-based text-to-image systems could not have been created without open sharing of software and data; and thus, we require that our youth participants (1) use text-to-image systems as part of their artistic toolbox, (2) view AI-generated images by prompts as open source images, and (3) create artworks themed on Mars Sample Return with open planetary images and Pixlr, a powerful open-source online graphic design tool.

Table 1: HiRISE+AI Curriculum

Unit 1: HiRISE Images

- (1) Geologic features of Martian surface and HiRISE images of Martian landscapes
- (2) Mars Sample Return Mission and related artists' concept artworks

Unit 2: Python Basics with Applications in HiRISE Image Processing

- (3) Jupyter Notebooks & Google Colab
- (4) Python basics: indentation, comments, variables, lists, for loops, functions, PIL library
- (5) Image processing Implementation on Colab to illustrate how computers see a digital image
- (6) Google Drive, Google Colab, Google's GPUs, GitHub, HuggingFace Hub
- (7) Creating a collage of butterfly craters by using Python for loop on Colab

Unit 3: Convolutional Neural Networks (ConvNets/CNNs) with Applications

- (8) Basics of Convnets: convolution and pooling operations, filters, layers of CNNs, why convolutions?
- (9) Implementation of Edge detection on Colab to showcase how convolution extracts features from an input
- (10) Implementation of CNN models on Colab to showcase how CNN models automatically learn features from input data through multiple building blocks and backpropagation
- (11) Generative Adversarial Networks (GANs), what they are and how they are trained
- (12) Training ProjectedGAN on Colab with custom HiRISE dataset and Google's GPUs
- (13) Introducing AI ethics through training ProjectedGAN and visual inspection of generated synthetic images

Unit 4: Creating Artworks with Leading-edge AI Tools and Pixlr

- (14) Using Pixlr one-click method and using Background Removal Colab notebook for background removal
 - (15) Creating artworks by using Neural Style Transfer Colab notebook with HiRISE images
 - (16) Upscaling the Beagle 2 Lander image (a lost Martian lander) with ESRGAN on Colab
 - (17) Creating artworks themed on Mars Sample Return by crafting prompts on DreamStudio, Midjourney
 - (18) Creating artworks themed on Mars Sample Return with Pixlr and Studio Diffusion Colab notebook
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We present our youths' artworks at the Art Gallery of the Clark County Library which draws a large group of youth and adult audiences (See Figure 1 below) and also our online gallery. Preliminary findings show that the progression of activities in the units of HiRISE+AI and the overall breadth and depth of the curriculum have proved to be suitable for implementation among high school youths. The

HiRISE+AI is highly engaging, and youths fully participate. After participation in the HiRISE+AI, youths gain (1) an intuitive understanding of how a CNN model automatically learn features from input data through backpropagation by using multiple layers; (2) a concrete experience of how ProjectedGAN generates realistic images; (3) insight on the inner workings of content-generation algorithms, and (4) a deeper understanding of AI ethics through visual inspection of generated realistic images during the training ProjectedGAN process with custom HiRISE dataset: if the data fed into the model is flawed from the start and then the AI model could be biased and discriminatory. Youths' attitudes toward AI and interest and excitement about learning AI are quite high. Students at Clark High in Las Vegas start their own HiRISE+AI club and they have been reaching out to over ten schools in southern NV to promote our HiRISE+AI curriculum. One student states in her feedback *"I have learned so much from HiRISE+AI. I learned about a lot of the things you can do with AI, machine learning, python, and HiRISE images! I have enjoyed trying different text prompts and sifting through many outputs at the start. I have created using Google Colab notebooks and I have had so much fun creating artworks with my favorites being: ProjectedGAN, Neural Style Transfer, and Stable Diffusion just to list a few! I want to learn more about programming and machine learning!"*

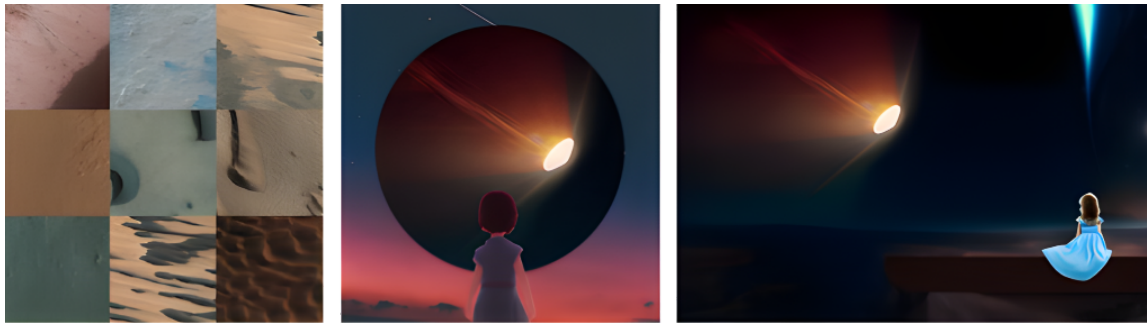


Figure 1: Youths' Artworks. Left: AI-generated Martian Landscapes, Middle and Right: Two Artworks Themed on Mars Sample Return

We are encouraged by the success of our pilot study. Aided by the insights from *HiRISE+AI*, we propose to develop and research a scaled-up project to engage underrepresented youths in East TN.

1.3 Broader Impacts

During our HiRISE+AI exhibits at the Clark County Library's art gallery, we were confronted with conspiracy theories on several accounts when we were explaining Mars explorations and the geologic features of the Martian surface. In order to dispute the conspiracies and avoid confrontation, we printed with us each time the statement by NASA spokesman, Allard Beutel, on a piece of paper: "a significant amount of evidence to support NASA landed 12 astronauts on the Moon from 1969-1972 is NASA has 842 pounds of astronaut-collected Moon rocks studied by scientists worldwide for decades; you can still bounce Earth-based lasers off the retroreflector mirrors placed on the lunar surface by the Apollo astronauts; NASA's Lunar Reconnaissance Orbiter imaged the landing sites in 2011." Youths growing up today in and out of the United States must learn to navigate a more complex world increasingly influenced by misinformation and disinformation about the STEM enterprise. Space disinformation erodes trust in science; and emerging AI, particularly machine learning and deep learning, is poised to amplify disinformation. However, AI also offers enormous promise to empower youths to address misinformation and disinformation (Bontridder and Poulet, 2021). Our proposed project here deliberately addresses this key issue by encouraging our youths to take on leading roles as creators of AI-driven technology to address space misinformation and disinformation through youth's own creative works which form connections between science and the public while illustrating planetary explorations and discoveries.

Our proposed work addresses multiple local and national needs. In addition to empowering youths to create artworks, web apps, and digital stories to tackle space disinformation from youths' perspectives, we will broaden underrepresented youths' access to emerging leading-edge AI technologies and include youths in the ongoing conversation in AI ethics and AI's contribution to creative works. East TN has experienced a lack of information access and use of information technology, and a lack of adequate education among the lay population and extreme conditions of information poverty (Crowther et

al., 1992; Mehra, 2010). We seek to address the local need by providing a locally relevant out-of-school learning experience which engages youths in leveraging recent availability of Earth observation datasets and leading AI tools and creating scientifically valid and visually appealing creative works. Through creation and dissemination of their creative works to respond to the issues and challenges locally youths will make a positive impact in their own communities. In this project, through creative production underrepresented youths are not simply AI-based technology consumers, they are the creators of AI-driven technologies, in addition to conscious users of emerging AI technologies.

Both educational and scientific research stemming from this project will be presented in conferences and published in relevant peer-reviewed journals. Generated open educational materials will be shared with schoolteachers, informal STEM educators, outreach professionals, and local communities via links to the project website, and will broaden access to high-quality interdisciplinary educational materials and improve the overall quality of teaching and learning. This program will leverage previous federal investments by making the state-of-the-art lab facilities at the University of Tennessee, Knoxville (UT) open to participating youths and educators, so that they can have hands-on experience with cutting-edge analytical facilities. Our proposed work will expose upwards of thousands of individuals from the local community to a brand-new learning experience.

1.4 Related Prior NSF Support

Co-PI Nicholas Dygert: NSF 2054551 [2021-2024] Co-Investigator-"Collaborative Research: Magmatic and Mechanical Extension of the Challenger Deep Forearc Segment: Insights into Subduction Initiation." Awarded Amount: \$124,192. Intellectual Merit: This project aims to study a modern example of subduction initiation in the Western Pacific near the Challenger Deep and Guam USA, using a research vessel and a tethered remotely-operated vehicle. Dygert is actively working to recruit a graduate student for 2023 admission so they can join in the shipboard work. Broader Impacts: The team of early career and senior geoscientists will provide training for new marine geoscientists. Robust internet connectivity will allow a handicapped geoscientist to participate in expeditionary science.

Co-PI Shichun Huang: EAR-1919061 [2019-2022] "MRI: Acquisition of an MC-ICP-MS for Earth, Environmental and Planetary Research at the University of Nevada, Las Vegas". PI with Co-PIs Ganqing Jiang and Matthew Lachniet. Awarded amount: \$675,292; AST-1910955 [2019-2023] "Modeling dust condensation in protoplanetary disks". Co-PI with PIs Jason Steffen and Zhaohuan Zhu. Awarded amount: \$552,474. Intellectual Merit includes development of a self-consistent model for disk formation and dust condensation, constraints on the properties of the molecular cloud core from which the Solar System formed, and insights into the effects of outbursts on the early thermal history of the Solar System. Ongoing work is investigating some aspects of variation in composition for the solar nebula. Broader Impacts: With NSF support, we designed and offered a summer analytical class (both in 2021 and 2022) to train students with cutting edge ICP-MS analytical techniques. As a result of our summer program of ICP-MS, a female student first authored abstract was presented at a local science conference in Feb 2022 [Lee et al. 2022], and an abstract first-authored by a high school student was submitted to 2022 AGU fall meeting [Jiang et al. 2022]. Outcomes: These grants led to 6 published papers, and 4 manuscripts under consideration.

Co-PI Pengyu Hong: OAC-1920147 [2019–2022] NSF MRI: Acquisition of a GPU-Accelerated Research Cluster; Amount of Support: \$349,983. Intellectual Merit: This award supports Brandeis to build a GPU cluster that will enable the Brandeis scientists to carry out big-data-driven research and education. The scientific questions span from the atom-level to groups of organisms and from quantum to cognitive. Broader Impacts: The proposed cluster will enable Brandeis to increase synergy between research and education at all levels and provide an environment for the future workforce to gain STEM experience. Outcomes: The GPU cluster is up and running and has led to 15 publications.

Co-PI Vasileios Maroulas: DMS 2012609 [2020–2023]: Quantum Computational Signal Classification; Amount of Support \$300,000. Intellectual Merit: This award develops a quantum topological framework which analyzes data from by relying on principles of quantum machine learning algorithms. Broader Impact: This is expanding the capacity for targeted signal detection in noisy domains. This award is also supporting one LatinX GRA. Outcomes: This project has led to 4 peer-reviewed publications thus far.

2. Project Design: Designing the STEM-based and Interdisciplinary Knox Innovation Lab

Our proposed work is designed around a three pronged approach: (1) utilizing an interdisciplinary approach integrating Planetary Science, big data, creative works, and AI, (2) building a Community of Practice (CoP) to facilitate youths' STEM learning and creative expression, and (3) Utilizing equity-oriented approaches to support the engagement and persistence of youths in educationally under-resourced East TN (See Table 2 Logic Model below).

Table 2 Logic Model

INPUTS	ACTIVITIES	OUTPUTS	SHORT-TERM OUTCOMES	LONG-TERM OUTCOMES
1) Expert knowledge in Planetary Science and AI 2) Expertise in developing, implementing and researching informal STEM programs 3) Building on experiences from multiple NSF funded programs 4) Building on a pilot study HiRISE+AI project 5) Successful existing partnerships with local communities, school districts and commitment to developing interdisciplinary STEM	<u>Planet+AI Art Challenge, App Challenge, Digital Storytelling Challenge</u> 1) Youths participate in the Challenge and are supported by researchers via CoP. 2) Challenge themes include local issues & challenges. <u>After-school Webinars/Workshops</u> 1) Youths will be facilitated by researchers & educators. 2) Webinars series by invited speakers <u>Summer Institute</u> Youths will participate in the program. <u>Planet+AI Art Exhibits</u> Online exhibits and physical Exhibits will be on display in museum-like public spaces <u>Innovation Conferences</u> Bringing youths, researchers and communities together. <u>Curriculum Materials</u> 1) Iterative development of curriculum enhancements for interdisciplinary STEM 2) Out-of-school interdisciplinary STEM	1) ~120 youths participate in the Challenges. 2) ~20 educators participate in PD workshop 3) Website, Slack, Rocket Chat, Google Drive, GitHub repos, social media are accessible, populated and visited.	1) Increase in youths' knowledge and skills in Planetary Science & AI 2) Increase in youths' science identity & sense of belongs to a scientific community 3) Increase in youths' enthusiasm in STEM & creative expression 4) Enhancement in youths' interdisciplinary problem solving and creative thinking	1) Established sustainable interdisciplinary program integrating Planetary Science & AI featuring professionally formatted educational resources 2) Materials are shared online and with educators at local and national or science teacher conferences. 3) Youths' persistence in and knowledge of STEM skills 4) Wide adoption and dissemination of generated educational resources

2.1 Utilizing an Interdisciplinary Approach

Through a 40-minute meeting with a group of target youths and educators to brainstorm ideas for this proposal, we have learned that our proposed project should be relevant and provide multiple entry points for youths. We have also learned that many youths may not be interested in science yet are definitely interested in producing creative works with AI. With that in mind, we use social constructivism to form the basis of our proposed interdisciplinary informal STEM project. Social Constructivism states that youths learn by building personal interpretations of the world based on their experiences and interactions with the environment (Stavredes, 2011; Ertmer & Newby, 2013). Therefore, youths are seen as active participants in the learning process (Ertmer & Newby, 2013). Science identity and belonging can be informed by social constructivism; as youths learn from and with others, they also begin to identify with an ingroup of the like-minded. Our proposed project will start by presenting high school youths with three Challenges which serve as the organizing center and context for learning (Bybee, 2010; Ashgar et al., 2012). These three challenges will be themed on planetary explorations and discoveries and other themes. Youths can also create their own themes. We will encourage youths to choose those themes related to the issues and challenges which are unique to the low-income and rural communities in East TN. Our Planet+AI Challenges will include (1) Planet+AI Art Challenge which engages youths in creating

meaningful aesthetic artworks by leveraging emerging AI's text-to-image systems, (2) Planet+AI Digital Storytelling Challenge which engages teams of youths in working creating multimedia-rich digital stories by utilizing emerging AI's content (texts/images/videos/audio/music) generation systems, and (3) Planet+AI Web App Challenge which engages teams of youths in creating web apps by utilizing AI's code generation tools and other content generation tools in order to respond to real-world issues impacting diverse communities. Many informal STEM programs show that adding arts to STEM helps to broaden participation in STEM learning (Conner et al, 2017), expand corresponding disciplinary literacies and workforces (e.g. Land, 2013), enhance creativity and self-expression (e.g. Schneider, et al, 2016), and thus prepare youths for the changing world and workforce (e.g., American Innovation and Competitiveness Act, 2017). Digital storytelling serves both as an effective tool for learning and also as a hook to capture the attention of youths and increase their interest in exploring new ideas, and help youths retain new knowledge as well as aid in the comprehension of difficult materials (Burmark, 2004; Ormrod, 2004; Hibbing & Rankin-Erikson, 2003; and Boster et al., 2002). PI Wang has been organizing NASA's Space Apps Hackathon in the past four years (see #2-4 in Reference Section) where we learn that youths are more motivated and encouraged when they maintain ownership over what they can study and investigate. By creating web apps to address local community issues and challenges youths will become creators of AI-driven technology, in addition to critical consumers of AI-based technologies. In addition, creating artworks, digital stories and web apps engage youths and adult audiences to think deeply about social, scientific, and technological issues from multiple youth perspectives.

We will display our youths' artworks which grapple with STEM content in our partner museums, including the AMSE, which can serve as an ideal informal learning context for both youths and adult audiences to explore scientific questions. Works of art can communicate programs and issues of a scientific nature in ways that words, numbers, and mathematical symbols cannot (Eisner, 2022). In contrast, the exhibitions of artwork that explores scientific content can offer viewers opportunities to see alternate, sensory-driven forms of communication and the expanse of ideas that follow (Madden et al., 2022). We use Quidley et al. 2017 conceptual model to design our instructional content and learning context. We will actively disseminate our youths' digital stories through Medium, NPR and East TN PBS. Youths will integrate arts with STEM in the following ways: (1) communication, or using arts to communicate planetary explorations (e.g., creating an artistic map about the cattle farming by using satellite image and Neural Style Transfer model); (2) creative expression, or displaying scientific information in novel or unique ways (e.g., creating an artistic collage of different types of craters on Mars); (3) content explanation, or using arts to explain scientific phenomena (e.g., using slow-motion video of how butterfly crater forms in digital storytelling); and (4) community building, or using arts to build relationships (e.g., starting youth's Planet+AI clubs).

2.2 Building a Community of Practice (CoP) to Facilitate Youths' STEM Learning and Creative Production for Real Audiences

Youth participants will be enrolled into a Planet+AI Community of Practice (CoP) (Lave & Wenger, 1991; Wenger, 1998;) when they register for either the Planet+AI Art Challenge, App Challenge, or Digital Storytelling Challenge in each of the three years of the proposed project. Educators will be recruited into the CoP by holding a week-long summer workshop in the final year of the proposed project. Scientific communities of practice have shown to be able to successfully introduce youths to scientific investigation practices, build their confidence in science, and sustain their participation. Our proposed 3-week summer institute each year, and the subsequent after-school webinar/workshop series, and 1-week PD workshop in the final year will be primarily staffed by the existing members of the CoP (Planetary scientists, AI researchers, experts in creative arts and members of the broader scientific and creative works community). The aim of the CoP is to facilitate and bolster students' STEM learning and expression. As AI technologies continue developing and being adopted, it's crucial to add the after-school webinar/workshop series in the Fall and Spring semesters in order to introduce our youths most recently open-sourced AI tools and current planetary explorations. Creativity is a systemic act of interaction between the creative person and its socio-cultural environment, which will recognize it as a genuinely creative act (Plucker et al., 2004; Guilera, 2011). We will manage our CoP actively and collaboratively (Fleith, 2019) in order for our CoP to offer support for the development of each youth's capacities and have his talents recognized and encouraged with the appropriate instruction, materials, and presentation

channels. During PI Wang's HiRISE+AI project, many Co-PIs of HiRISE have been supporting our youths by giving Zoom presentations and answering our youths' questions via emails. Our youths were particularly inspired by Dr. Randolph Kirk's words on his achievements "Perseverance was the ninth Mars rover to go to a site we mapped, all of which landed successfully. Starting in 1962. Ten Mars landings attempted without USGS mapping failed for a variety of reasons. Perseverance also carried our maps onboard and used them to guide its final descent, a technological first." Encouraged by the HiRISE+AI, we will also conduct after-school webinar/workshop series during the Fall and Spring semesters in the proposed project in order for our CoP to encourage our youths' STEM learning and creativity expression.

2.3 Using Equity-oriented Approaches to Support the Engagement and Persistence of Youth Participants

Our proposed work will take an assets-oriented perspective to build and value "funds of knowledge" of diverse students and offer students with space to make sense of science, science people, and their current and future selves based on their experiences in school and after-school science contexts. Incorporating a "strengths-based model" entails youth as having capital in the form of success and persistence. The Community Cultural Wealth model (Yosso, 2005) describes youth as having capital in the form of "community cultural wealth". Through this lens, youths are seen as bringing six kinds of capital to STEM programs: aspirational capital, linguistic capital, familial capital, social capital, navigational capital, and resistant capital. These ideas have been applied to STEM studies of informal education (e.g., Aragon, 2018; Kelly et al., 2019) and will be integrated into the proposed project so youths can leverage this capital within the program and as they move beyond high school. We will collaboratively develop the project's Challenge themes, Challenge Manuals, activities, and curriculum materials with Planetary Scientists, AI researchers, educators, youths and their parents, and community members in consultation with local school district's science curriculum experts (including Dr. Deanna Pickel from Oak Ridge High School and Ms. Melody Hawkins from Austin-East High School) to ensure the project is creating an inclusive and engaging environment for youth. Our partner school, Austin-East High School has a 88.5% minority enrollment in Knox County (see #5 in Reference Section). Sunbright School is a rural school and serves as the community center in the area and it has 69% of students on free and reduced lunch assistance (see #6 in Reference Section). Youths' "community cultural wealth", particularly their aspirational capital (the ability to persist or to maintain resilience while in the pursuit of long-term goals), social capital (the social networks, previous experiences, and community resources accessed to create positive outcomes), and familial capital (the cultural knowledge and support accrued from family and extended family networks through validation, advice, and shared community history) will be leveraged in the proposed project. We will create opportunities for youths to share their accomplishments at the end of each program celebration. We will conduct Planet+AI art exhibits for youths at our partner museums and other venues. We will also build on youths' own experience with Planetary Science, AI and creative works in everyday life (such as in Google translation tools, Netflix, Instagram, etc.) and connect their learning to their futures. Through these mechanisms, youths receive validation of their funds of knowledge, reinforcement of their own STEM potentials, and are afforded the opportunity to imagine the realm of possible selves based on the skills they have developed. Further, we plan to systematically identify and work to mitigate disparities associated with usage of tech tools and activities in the project. Upon registration we will survey youths on their technology resources (computer, operating system, bandwidth, web browser). We will also administer exit tickets at the end of each session during the project in Year 1 to ask youths which activities are difficult to engage with due to technology limitations. Equitable strategies for engagement include modifications made to activities and tools to make them suitable for online settings with low-bandwidth and with older computers. For example, through previous experience we found that utilizing Zoom breakout rooms in which youths instruct facilitators to run activities on facilitators' machines while screen sharing so youths can view the outcomes as a mitigation strategy for youths with low difficulty running the applications on their own machines.

2.4 Impact on STEM outcomes

We expect that the Knox Innovation Lab will increase youths' knowledge and skills in Planetary Sciences & AI, and their motivation to stay engaged in STEM. As youths respond to the Lab's challenges, they will not only learn how to explore planetary images and coding, but will gain an enduring understanding of several "Big Ideas" in AI, including how computers use algorithms to learn from data,

and how “AI can impact society in both positive and negative ways” (AI4K12, 2022). Creating media about and with AI has taught high school youth the basics of how AI works, and empowered them to become critical consumers of the AI they encounter in their lives (Lee et al., 2022). We also expect that the proposed project will affect youths’ social-emotional learning, especially their STEM identity and sense of belonging to a scientific community. In keeping with Kim, Sinatra, & Seyranian (2018), we define STEM identity as “a socially based identity grounded in the extent to which individuals see themselves and are accepted as a member of a STEM discipline or field.” (p. 591). In other words, identity is shaped in part by how individuals relate to varying group environments. Belonging, or being part of an ingroup, is an “inference, drawn from cues, events, experiences, and relationships, about the quality of fit or potential fit between oneself and a setting. It is experienced as a feeling of being accepted, included, respected in, and contributing to a setting, or anticipating the likelihood of developing this feeling.” (Walton & Brady, 2017, p. 272). Peers, near-peers, and adults all have a role in creating an environment that can support youths’ sense of belonging. This shapes youths’ identities as “science people.” It is especially important to recognize youths’ cultural funds of knowledge to facilitate belonging in out-of-school programs (Nixon et al., 2021). In East TN, students have fewer opportunities to interact with others who are interested in STEM, and to recognize how their STEM knowledge and skills are valuable to local and global contexts (Coggin, 2017; Saw & Ager, 2021). The Knox Innovation Lab is uniquely positioned to empower youths to become lifelong, creative scientific learners whose assets enhance their communities by creating STEM-rich artworks, digital stories and web apps that respond to issues and challenges globally and locally and utilizing AI tools innovatively.

3 Project Implementation: Building Capacity to Sustain and Grow Knox Innovation Lab

3.1 Project Components

3.1.1 Planet+AI Art Challenge, Digital Storytelling Challenge and App Challenge

The core activities of our proposed work are a set of three Challenges, the Planet+AI Art Challenge, Digital Storytelling Challenge, and App Challenge which engage youths in creating scientifically valid and visually appealing creative works by leveraging leading-edge AI tools. The Challenge themes will be centered on planetary explorations and discoveries, or addressing some issues and challenges unique in local communities. Youths can choose any of these themes and can also create their own themes. We will launch the three Challenges in January each year.

3.1.2 Planet+AI Curriculum

Building on the success of PI Wang’s HiRISE+AI, we will design and implement the Planet+AI curriculum for high school youths with no prior knowledge or skills in coding or AI. For the AI trajectory, our Planet+AI curriculum will be based on the following theoretical foundations and recent research in AI education: (1) The Computer Science Teachers Association revised 2017 Standards and included two standards at high school level that explicitly address AI (CSTA, 2017). (2) The AI4K12 Initiative has published guidelines for teaching AI in grades K-12 organized into 5 big ideas. (3) Recent studies have been focusing on AI literacy (e.g., Druga et al., 2019; Cheng et al., 2020; DiPaola et al., 2020; Lee & Ali, 2021; Long & Magerko, 2020; Greenwald et al., 2021) and fundamental AI concepts (e.g., Ali et al., 2019; Touretzky et al., 2019; Williams et al., 2019; Sabuncuoglu, 2020; Lee et al., 2021).

We will continue using the same application-focused approach and the same progression of activities in our pilot HiRISE+AI curriculum. During their creative process, youths will be required to use AI tools as part of their artistic toolbox and view AI-generated content from prompts as open source content. We have had a series of consultations with mathematicians and computer scientists (including Dr. Fred Huffer at Florida State University and Dr. John Dongarra, the 2021 Turing Award winner) on linear algebra basics and probability distribution basics, and also high school teachers (including Ms. Melody Hawkins). We anticipate that there will be more advanced and more powerful open-source AI tools in the coming years. Considering what works and what needs to be upgraded on HiRISE+AI for the proposed scaled-up Planet+AI curriculum, we will (1) add linear algebra basics and probability distributions basics for convolutional neural networks, (2) introduce current planetary explorations and update leading-edge Generative AI tools, and (3) use language model example as well as ProjectedGAN-training to discuss AI ethics (see Table 3-4 below).

Table 3: Planet+AI Curriculum**Unit 1: Planetary Images**

- (1) Introducing images from previous and current planetary explorations, Earth observation images, and the Scanning Electron Microscopic (SEM) images of meteorites
- (2) Planetary exploration missions and related artists' concept artworks themed on the missions

Unit 2: Python Basics with Applications

- (3) -(7): same as (3)-(7) in HiRISE+AI Curriculum

Unit 3: Convolutional Neural Networks (ConvNets/CNNs) with Applications in Planetary Image Analysis

- (8) Operations on the digital planetary image using notation and operations of linear algebra
- (9) Implementation of probability distributions simulations on Colab for Intuitive understanding of probability
- (10)-(14): same as (8)-(12) in HiRISE+AI Curriculum

Unit 4: Creating Artworks with AI Tools and Pixlr

- (15)-(19): same as (14)-(18) in HiRISE+AI Curriculum

Unit 5: Creating Artworks, Digital Stories and Web Apps with Emerging Leading-edge AI Tools

- (20) Text generation by crafting prompts with GPT-3
- (21) Python Code generation by crafting prompts with Codex
- (22) Building a "Name My Team" web app using OpenAI's API and language model
- (23) Other content generation with emerging leading-edge AI tools
- (24) Fine tuning on a small planetary image dataset for text-to-image models and other models

Unit 6: AI Ethics:

- (25) Bias in dataset through training ProjectedGAN and visual inspection of generated synthetic images
- (26) Language models can have a pernicious tendency to reflect societal biases.

Table 4: Principles for Designing Planet+AI instruction**Instructors:**

- (1) Instructors take the role of facilitators.
- (2) Planetary Scientists, AI researchers and artists give Zoom webinars and answer youths' questions.

Course Design on integrated Planet+AI and also on soft skills:

- (3) Our Planetary Scientists introduce planetary science basics and planetary explorations.
- (4) We use Participatory Live Coding and "use-modify-create" throughout the project.
- (5) We enable youths to create their own independent projects, but offer regular feedback.
- (6) We encourage youths to share technical strategies and discoveries
- (7) We develop youths' 21st century skills including creativity, critical thinking, problem solving, collaboration.

Youths Engagement and Retention:

- (8) We Recruit youths from different backgrounds but with common interests.
- (9) We use Google Drive, Slack, GitHub which permit instructors to provide individualized attention.
- (10) We conduct after-school webinar/workshop series to introduce most recently open-sourced AI models and current planetary explorations and use these great achievements to engage youths in STEM learning.
- (11) We conduct youths' artwork exhibits both online and also in partner museums.

We anticipate that after participation in our Planet+AI, youths will (1) gain insight on the inner workings of emerging content-generation algorithms based on an intuitive understanding of convolutional neural networks and training ProjectedGAN with custom HiRISE dataset, (2) develop skills in crafting prompts and fine-tuning to generate content (texts/images/music/videos/code) with leading-edge generative systems, (3) have a general idea about the general capabilities of AI in our time and how deep learning methods affect the society and individuals, and (4) critically review AI ethics.

3.1.2 After-school Webinar/Workshop Series

As AI is constantly developing, we will introduce the most recently open-sourced Generative systems and current planetary explorations and discoveries to our youths through after-school webinar series by invited speakers in Fall and Spring semesters. Keeping up with emerging technologies is one of the most exciting aspects of PI Wang's current HiRISE+AI project, which effectively contributes to the engagement and persistence of youths' participation in our academic year-long program. Our after-school webinar series will cover topics including current planetary exploration and discoveries, emerging AI's content generation tools, and addressing local issues and challenges unique in low-income and rural communities in East TN. Our workshops will be conducted by our project staff to facilitate youths to finish

their Challenge projects. Our project staff will answer youths' questions via emails, Slack, Google Drive, etc and offer feedback throughout the project.

3.1.3 Summer Intensive Institute to Integrate STEM into Artworks, Digital Stories and Technology Production

We will implement the following activities and explorations over the three-week period:

Scientific Practices: Youths will engage in a series of theme-related science activities where practices of science are made explicit. For example, if the theme becomes impact craters, the youth will participate in crater-formation experiments and a field trip to the cutting-edge lab on the campus of the UT. Through reflection on those practices, youths will examine what concepts were explored in their experiments and how the practices used in those experiments can be represented in their artworks, digital stories and web apps.

Planet+AI: Graduate students and undergrads of Computer Science major from Brandeis Univ. and those of Planetary Science major from UT will be our instructors (see Figure 2-4 for examples of Planet+AI Curriculum).

Daily Activities: We will structure the daily activities around a suite of principles that are designed to engage youths in connecting their personal experiences to create artworks, digital stories and web apps that are of interest and value to them and their communities. Our everyday activities will utilize Putting Learning By Design framework (Kolodner et al., 2002) and follow a similar structure, including (1) warm-up activities, (2) Science Time: Connecting to Personal Experience, (3) Artworks, Digital Storytelling and Web App Time, (4) Discovery Time, (5) Synthesis Time, and (6) Reflection Time.

3.1.4 Planet+AI Art Exhibits and Innovation Conference

Presenting science is an important part of doing science and is also very rewarding for young investigators. As mentioned above, it is also an important part of claiming one's STEM identity as a member of the science community (Gee, 2000). Our Planetary scientists and project staff will ensure all our youths' artworks, digital stories and web apps are scientifically valid. We will create opportunities for youths to share their accomplishments throughout the project with a variety of options. We will host an online Planet+AI Art Exhibit throughout the project. We will also organize physical Planet+AI art exhibits at our partner museums including the AMSE. Our youth artists will explain their artworks in front of their exhibits, or we will show their recorded videos by the side of their artworks. During the art exhibits at our partner museums, our STEM professionals and facilitators will communicate with the public audiences by the side of our youth artists and story-tellers on Planetary Science and AI and answer their questions. We will also bring meteorite/mineral kits, computers with prepared Colabs and other tools and resources needed to the site so teens and adult audiences are able to create artworks right away on the site. Our trained facilitators will explain the science behind the artworks. When explaining the SEM images of meteorites, our facilitators will bring meteorites to the Museums with the exhibits. We will continue to hold youths' Planet+AI artworks at the Science and Technology Festival, the Smithsonian Learning Lab, the Annual Knox Asian Festival, Annual HoLa Festival, and Black History Month events. At the end of the Planet+AI App Challenge each year, we will host an Innovation Conference via Rocket Chat where students will present their Challenge projects to their families, friends, and local community members. PI Wang has been using Rocket Chat for Space Apps Challenge over the past few years and it has been proven to be safe and effective.

3.1.5 One-week PD Workshop in the Final Year

We will hold a one-week workshop in the final year which will train educators to use our learning materials in either formal or informal learning settings. This one-week workshop will focus on (1) crafting

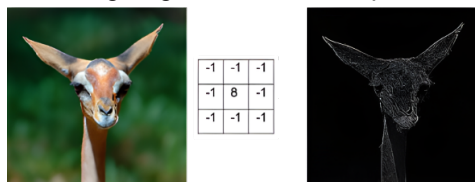


Figure 2: Convolution Operation: Edge Detection

```
from google.colab import drive
drive.mount('/content/drive')

from PIL import Image, ImageFilter
image = Image.open('/content/drive/MyDrive/deer.png')
image = image.convert('L')
edge = image.filter(ImageFilter.Kernel((3, 3),
(-1, -1, -1, -1, 8, -1, -1, -1, -1), 1, 0))
edge.save('/content/drive/MyDrive/edge_deer.png')
```

Figure 3: Python Script for Edge Detection implemented on Colab

Name My Team

Knoxville AI Warriors

Figure 4: A "Name our Knoxville Team" Web App by Our Student

prompts to generate content with leading-edge AI's content generation systems, and (2) creating artworks, digital stories and web apps by utilizing emerging AI's generative systems (e.g., text-to-image algorithms such as Midjourney) as part of the artistic toolbox and viewing AI-generated content from prompts as open source images. We will actively get educators' feedback and the elements of the program they would like to be able to do with their students.

3.2 Project Implementation

3.2.1 Recruitment with Our Partners

Student Recruitment: Through collaboration with the UACS and AMSE we will visit our partner schools during periods for a regularly scheduled school assembly. We will also promote our Planet+AI Art Challenge, App Challenge and Digital Storytelling Challenge at the school's Parent's Night. We will promote our programs at our partner museums by showcasing our Planet+AI artworks, and also by bringing well-designed hardware and software so youths and adults can easily create artworks and right away on-site. Our community engagement and outreach specialists, Dr. Amelia Robinson and Dr. Shalaunda Reeves, and our collaborators, Dr. Janine Al-Aseer and Ms. Melody Hawkins, will be the role models for girls and those from underrepresented minority groups. We will anticipate that we will recruit about 120 students in the three-year project. We will design a series of themes including beautiful Tennessee and Smoky Mountains in East TN, so youths can use satellite images and AI algorithms to create artistic maps or collages. Also, our researchers and educators will work together with parents and community members, particularly those in rural Morgan County, to design a series of themes based on the local community's needs and interests. We will create visual documentation panels and web-based digital stories to communicate process and progress. Shared experiences, particularly if embedded in a non-traditional STEM learning experience, between parents and teens within the STEM domains have the potential to increase youth interest and confidence toward science (Syed et al, 2011). To that end, we will develop materials that will go home to parents, including short descriptions of the STEM topics that youth are working on as a part of their projects, and describing how engagement in artworks, digital stories, and web apps production support youth in applying STEM content knowledge and to develop skills such as collaboration and presentation skills.

Educator Recruitment: In collaboration with the UACS, our partner schools and partner museums, about 20 educators (including schoolteachers and informal educators) will be recruited. Priority will be given to those from underrepresented groups.

3.2.3 Research Plan

Staff from Rockman et al Cooperative (REA), an independent research and evaluation company, will conduct the project's research. They will use mixed methods to investigate the following research questions: (1) How does participation in an interdisciplinary out-of-school STEM program influence youths' AI knowledge and skills, STEM engagement and identity, and sense of belonging to a scientific community? (2) What do youths learn about the value of their personal and cultural knowledge and skills (i.e., assets) to the local and broader community? (3) What program Challenges, features, and content appear to have the greatest influence on engagement, identity, and belonging? (4) How do outcomes vary by youth demographics (e.g., gender, ethnicity, urban/rural environment)? The questions test the project's theoretical assumptions that programming grounded in a socially constructed community of practice which recognizes youths' community cultural wealth will facilitate sustained STEM engagement.

Sample and general approach. Researchers will work with all youths who consent to participate in the study (maximum N = 120). They will pilot the study instruments and procedures in Year 1 before applying a finalized methodology in Years 2 and 3. To ensure that the study is culturally responsive (Bryan & Lewis, 2019), they will vet the instruments with advisors, youths, or members of the project team who are the most familiar with the project's educational and community context. They will member-check their findings and interpretations through activities such as data parties (Franz, 2013). Researchers will approach their work with cultural humility (Haynes-Mendez & Engelsmeier, 2020), and correct identified misconceptions and inequities in methods and results.

Instruments. All youths will complete annual pre and post surveys on their STEM engagement (Student Interest in Science, Technology and Geospatial Technology: Bodzin et al., 2020; STEM Learning Activation Survey: Moore et al., 2013), identity (Hill et al., 2018) and belonging (Classroom Community Scale: Rovai, 2002; Student Measure of Culturally Responsive Teaching: Dickson, Chun, & Fernandez,

2016). When necessary, researchers will adjust item wording to fit the Knox Innovation Lab context. Their AI knowledge and skills will be assessed with a checklist and rubric for their final project (Cateté, Lytle & Barnes, 2018), and a survey about AI technologies and ethics issues, relevance of AI, and awareness of AI impact on future jobs (Zhang, et al., in review). The latter measure was funded by NSF, and developed and validated with middle and high school youth of diverse backgrounds.

Youths will also create written or video recorded youth artists' statements (Bass & Bandy, 2009; Bass, Hu Dahl, & Panahandeh, 2016) that they can also display at the public exhibitions. They will be asked to share (a) their inspiration for their STEM-rich and scientifically-valid artworks, web apps, or digital stories, (b) the message(s) they wanted to communicate, (c) the ways their work reflected themselves and their communities, (d) the challenges they encountered and how they resolved them, and (e) the nature of their community collaboration (e.g., who they worked with and in what roles).

Each year, ten youths will also be asked to document their experiences using Photovoice, a participatory research method (Wang & Burris, 1997). Researchers will sample youth from diverse racial, ethnic, and socioeconomic backgrounds, as well as across the program Challenges and locations (urban, rural). Youths will take or select pictures or short videos in response to themes or prompts. They will also describe the images. For example, one prompt will be "Please take and select 1-3 pictures that show what you think (a) it means to do science in your community; and (b) when and where you do science in your community." Other questions will be "What is AI? How would you explain it to someone your age?" "When do you use AI in your life?" During a sharing session, a facilitator will probe for details (e.g., why youth selected an image, what was significant about the event or individual depicted) using a semi-structured protocol, and ask the youth about themes they see across the images.

Analysis and reporting. Researchers will analyze all data with quantitative and qualitative techniques appropriate for the data collected to document how, for whom, and in what contexts the art-inclusive, interdisciplinary program facilitates knowledge, engagement, identity, belonging, and awareness of community-valued knowledge and skills. For example, researchers will use repeated-measures analysis of variance with between subjects factors to compare pre-post survey responses by program Challenge and location. They will correlate scaled scores of AI awareness with the STEM socio-cultural measures. Researchers will use Grounded Theory (Strauss & Corbin, 1990) to track the most common program features (e.g., art, AI, coding, community, public exhibits) and assets in youth artists' statements and Photovoice images. They will also build off of prior NSF-funded research on AI youth media production to code for evidence of computational thinking (e.g., technical considerations and problem-solving) and creative expression (e.g., aesthetic decisions and application of personal preferences and assets) (Lee, Gobir, Gurn, & Soep, 2022). Analyses from multiple data sources will be synthesized into broader themes.

4. Evaluation

Evaluation to support internal improvement. The project team will utilize a design-based research approach (Anderson & Shattuck, 2012; Design-Based Research Collective, 2003) to incorporate youths and educators' feedback into the design and refinement of the program to ensure the project is engaging and meaningful to underrepresented students. We will engage one iterative cycle of design and development per grant year. In Year 1, we will focus on designing and pilot testing the curriculum and activities. We will solicit feedback from youth through interviews and open-ended questions in surveys and exit tickets (i.e., whether/how engaging the activities are, how to improve the activities) and from teachers for content quality and usability (e.g., consistent language, whether the program is appropriate for the target audience). We will coordinate data collection with the research team to minimize the burden on participants. In some cases, this may mean that the teams will use the same data, but analyze for different purposes (program improvement versus generating knowledge).

Evaluation to promote accountability. A six-person Advisory Board (See Table 5 below) will meet annually to examine the following questions: (1) What is the scientific integrity and technical quality of the proposed technologies and materials? (2) To what extent has the project provided sufficient support for the targeted rural youths and educators to engage in integrated STEM to strengthen their science inquiry with AI? (3) To what extent has the project attained the research and development objectives? (4) How well has the project embodied equitable practices for collaboration? The project and research teams will prepare a portfolio of key grant products such as program materials, an overview of the research

methods, and a selection of findings and community feedback. They will also share brief reflections on the collaborative process for creating these materials including key decisions and unanticipated issues. Advisory Board members will spend one day reviewing these materials with project staff via videoconferences. The project staff, on an as-needed basis, will seek advice from individual advisors on different components of the project based on their expertise. In total each member is expected to spend two days per year on these efforts. The chair, Dr. Kristin Bass, will spend one more day per year consolidating feedback and insights from all members and helping project staff to incorporate them into annual reports for NSF. The PI will meet with the research team to determine how to incorporate the Advisory Board's recommendations into the project's ongoing management.

Table 5: Advisory Board Members

NAME	INSTITUTION & POSITION	EXPERTISE
Kristin Bass	Rockman et al Cooperative	Educational Research
Atlas Wang	Assistant Prof. of Computer Science at Univ of Texas at Austin	AI
Yingjuan Ma	Researcher of College of Earth, Planetary and Space Sciences, UCLA	Planetary Science
Qiusheng Wu	Assistant Prof. of Department of Geography at UT	Open Planetary Data
Janine Al-Aseer	Clinical Assistant Prof. of Department of Theory & Practice at UT	Community Partnerships
Melody Hawkins	Science Teacher and Administrator at Austin East High School	Teaching, Administration

5. Communication and Sustainability

5.1 Sustainability

We will build upon our existing strategies and relationships that our team has developed during the past years to sustain and grow the work we proposed here. Immediately upon the initiation the project team will start investigating additional sources of funding. As the project develops a track record, we will seek funding from other federal agencies as well as private donors. We will actively seek support from: (1) NASA Space Grant Consortium at UT, because we can easily broaden our focus to include planetary explorations and massive satellite data of Earth; and (2) other federal agencies such as the DOE and DOD that have programs aimed at increasing the STEM workforce. Further, our proposed work has a built-in sustainability model. For the project to maintain the professional learning community as described in the project descriptions, the project will seek funding from the resources mentioned above. Educational materials developed will be available on the project website permanently and advertised by links on the UT, and Brandeis' websites.

5.2 Dissemination of the Work

In addition to media outreach, online AI Art exhibits, and physical AI Art exhibits at museums, the project website will advertise our proposed program. Our partner museum, the AMSE is widely advertised and well attended and will reach a large group of general audiences. We anticipate the Exhibits will attract media attention, which will help to communicate our work to a more general population. The project team will be presenting the outcomes of our research work at typical research conferences such as American Educational Research Association, the Lunar and Planetary Science Conference, the National Association for Research in Science Teaching annual meeting, and Association for the Advancement of Artificial Intelligence conference.

Moreover, researchers from Rockman et al (REA) will develop a communication plan jointly with the PIs, project team members, and community representatives. This plan will be adapted throughout the project. Efforts will focus on reaching three audiences: (a) Informal STEM educational researchers. The project team will collaborate on a publication for the Journal of STEM Outreach (JSO), an open access, peer-reviewed journal for researchers and practitioners who develop, implement, and evaluate projects that connect scientists with formal and informal educators; (b) Youths who participate in the engagement program. The project team will support youths who wish to publish their outreach experiences as a blog or article; (c) Public audiences. REA will tap project partners and community members to identify culturally appropriate and respectful methods of and platforms for disseminating interim and final research and evaluation results to the public. REA has experience developing short research briefs and blog articles for public audiences in conjunction with other NSF funded AISLs. These blog articles are available through popular online venues, including Medium, LinkedIn, and the websites of our partnering organizations.

6. Project Management

6.1 Project Expertise, Key Personnel, and Partner Roles

Our team is well suited to the task of creating and successfully managing the project. The interdisciplinary team has broad experience in informal STEM and outreach, including members who are primarily scientists, primarily educators, and Community Engagement and Outreach Specialists .

PI, Ping Wang, is a Researcher at the Department of Earth and Planetary Sciences, Univ. of Tennessee, Knoxville. She has significant expertise and experiences in managing partnerships and organizing youth competitions and Challenges. She has hosted high-quality STEM programs including the ROADS on Mars Challenge in 2018-2019 and organized the Space Apps Challenge as the Local Lead in Las Vegas in 2019, 2020, and 2021, and in Knoxville in 2022. As an informal educator, she has hosted dozens of webinar/workshop series by working with researchers and professionals in Planetary Science and AI in the past few years, which drew a large group of high school youths to use NASA's open planetary data and AI algorithms to respond to real-world problems locally and globally. She is the PI of the NASA-funded STEM program on Engaging Youth in Space Exploration with Imagery which is a pilot study for this proposed work. She will be the overall lead for this project. **Dr. Jeffrey Moersch**, is a Professor of Planetary Science at the Department of Earth and Planetary Sciences, Univ. of Tennessee, Knoxville. He will provide strategic guidance on Martian geology and remote sensing. **Dr. Nicholas Dygert**, is an Assistant Professor at the Department of Earth and Planetary Sciences, Univ. of Tennessee, Knoxville, and also the Director of the Univ. of Tennessee NASA Space Grant Consortium. He will provide strategic guidance on Planetary Science. He will lead the Lab visit. **Dr. Shichun Huang**, is an Associate Professor at the Department of Earth and Planetary Sciences, UT. He studies Earth's mantle and the early solar system. He will provide strategic guidance on Earth Science and the Solar System. **Dr. Vasileios Maroulas**, is a Professor of Mathematics at the Department of Mathematics, UT. His research portfolio engages computational probability, statistics and machine learning with computational topology and geometry for addressing interdisciplinary problems in data science and engineering. He will collaborate with other team members in developing statistics and mathematics components of the proposed learning materials and co-teach it. **Dr. Pengyu Hong**, is a Professor of Computer Science at Brandeis University. His research focuses on machine learning and its interdisciplinary applications. The data types in his research are heterogeneous and include images/videos. He will collaborate with other team members in developing the AI components of the proposed curriculum and other learning materials and co-teach it. **Dr. Amelia Robinson**, is a lecturer at the Department of Earth and Planetary Sciences, UT. She will collaborate with other team members in developing Planetary Science components of the proposed learning materials and co-teach it. She will also collaborate with other team members in community engagement and outreach. **Dr. Shalaunda Reeves**, is an Assistant Professor of STEM Education at the UT. She will be our outreach specialist. She will collaborate with other team members in reaching out to local underrepresented youths and educators in East TN. **Learning Scientist, Dr. Kristin Bass**, from Rockman et al Cooperative has a PhD in Education and Psychology from the University of Michigan where she studied project-based learning and assessment in K-12 science. She has over 20 years of experience in teaching and educational research, including time as an informal middle school educator. She will lead the project's research and provide data to the Advisory Board for project accountability. She will also be the Advisory Board Chair.

We have a core set of **community partners** who have the capacity and experience to implement and sustain the program over time, including the AMSE, UACS, and our partner schools. The **University-Assisted Community Schools (UACS)** has been serving for at least twelve years economically disadvantaged students in urban Knox county as well as in rural Morgan County through providing access to needed resources and out-of-school learning opportunities, which provides a natural opportunity to engage youths and their parents in learning STEM. We will utilize UACS' existing extensive experience and out-of-school programming for the proposed project. **Dr. Janine Al-Aeer** is the Director of the UACS and also a Clinical Assistant Professor in the Department of Theory and Practice in Teacher Education at UT. She will work with our outreach specialists Dr. Amelia Robinson and Dr. Shalaunda Reeves, and other project team members to ensure that our community engagement and outreach will succeed. Our partner museum, **the American Museum of Science and Energy (AMSE)**, is a Smithsonian Affiliate and has been serving the community since 1949 through exhibits, educational initiatives and public programming. **Quinn Argall** is the Collections & Exhibits Director at AMSE, and he will use his expertise in creative works and give our youths strategic guidance on creative process and

creative productions. He will also work with the project team to ensure that our Planet+AI art exhibits will be successful. **Ms. Melody Hawkins** is a science teacher and also an Administrator of Austin-East High School. She will use her expertise in Science Curriculum and instruction to ensure that our project will be equitable. **Ms. Tammi Campbell** is the Executive Principal of Austin-East High School. She will support our project team to reach out to underrepresented youths, particularly those from minority groups in urban Knox County. By leveraging our partnerships, we anticipate that the goals we have set out to achieve here are quite achievable.

6.3 Management Structure

To manage our work, we will form teams with overlapping project staff. We will form a Leadership Team which will consist of PIs including the administrators of partner schools, two teachers in the program, and, as the project progresses, two experienced youths. The Leadership Team will meet four times per year with the goals of examining the progress of the work, examining the researcher team's data and the internal formative evaluative research, and evaluating the future directions of the project. The Research Team will consist of project staff lead by PI Wang and Kristin Bass, which will focus on answering our research questions and moving the research agenda forward. The Development Team will be led by PI Wang and include Hong, Moersch, Dygert, and Huang. The Development Team will focus on creating the overall structure of the project. We recruited a wide range of members to serve on our Advisory Board. The diverse Advisory Board will provide us guidance and suggestions on how to improve the program leveraging their particular areas of expertise.

6.3 Timeline

Table 6 Project Timeline

September-December 2023: Building Capacity and Starting Exhibits & Webinar/Workshop Series (Year 1)

- GRAs and PI plan after-school webinars/workshops and summer institute and advertise to partner schools.
- GRAs and PI develop websites, Challenge themes, curriculum materials.
- GRAs and PI recruit undergraduates and train undergraduate facilitators.
- GRAs and PIs conduct Fall semester after-school webinar/workshop series.
- GRAs and PI conduct program alumni's Planet+AI art exhibits at partner museums and the online gallery.
- REA plans research program in consultation with the project team and community members.

Spring 2024, Academic Year

- GRAs and PI Launch Planet+AI Art Challenge, Digital Storytelling Challenge and Web App Challenge.
- GRAs and PI Recruit students at partner schools through collaboration with partner museums and UACS.
- GRAs and PIs implement Spring after-school webinar/workshop series.
- GRAs and PIs conduct Planet+AI art exhibits, festivals, and disseminate youths' digital stories.
- REA develops and pilots tests research instruments.

Summer 2024

- Conduct intensive Summer Institute, led by PIs, GRAs with participation of all project personnel.
- GRAs and PI conduct new exhibits at partner museums and disseminate youths' digital stories.
- REA conducts research. The Advisory Board conducts its formative evaluation. PIs and senior personnel meet to consider assessment report and make adjustments. GRAs update website with new materials.

Fall 2024, Academic Year (Year 2)

- GRAs and PIs conduct Fall semester after-school webinar/workshop series.
- GRAs and PI conduct exhibits at partner museums and the online gallery, disseminate youths' digital stories.
- Youths present their work at the Innovation Conference in December.
- PI and REA disseminates initial results of data collected and educational impacts of the project.

Spring 2025, Academic Year

- GRAs and PI conduct the same activities as in Spring 2024.
- REA revises research plan and instruments.

Summer 2025

- Same as in Summer 2024

Fall 2025 (Year 3), Spring 2026

- Same as in Fall 2024 and Spring 2025

Summer 2026

- Conduct intensive Summer Institute, led by PIs, GRAs with participation of all project personnel.
 - Conduct a one-week PD workshop for educators.
 - GRAs and PI conduct new exhibits at partner museums and disseminate youths' digital stories.
 - REA writes the final research report and PI writes the final NSF report.
-

References

The Reference section has the following sub sections:

Section 1: URL Addresses

Section 2: References Associated with Prior Support Section

Section 3: References Associated with Proposal Narrative

Section 1: URL Addresses

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